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Applications of LLMs

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Agenda

- **Agenda:**
  - The Transition from Foundation Models to Applications
  - Core NLP and Text-Based Applications
  - Domain-Specific Transformations (Code, Healthcare, Law)
  - LLMs as Autonomous Agents (Tool Use and Reasoning)
  - Deployment Challenges and Ethical Considerations
- **Learning Outcomes:**
  - **Map** core LLM capabilities to specific real-world industry use cases.
  - **Evaluate** the appropriate adaptation strategy (Prompting vs. RAG vs. Fine-Tuning) for different applications.
  - **Analyze** the architecture of LLM agents and tool-augmented systems.
  - **Critique** current limitations regarding deployment, cost, and safety.

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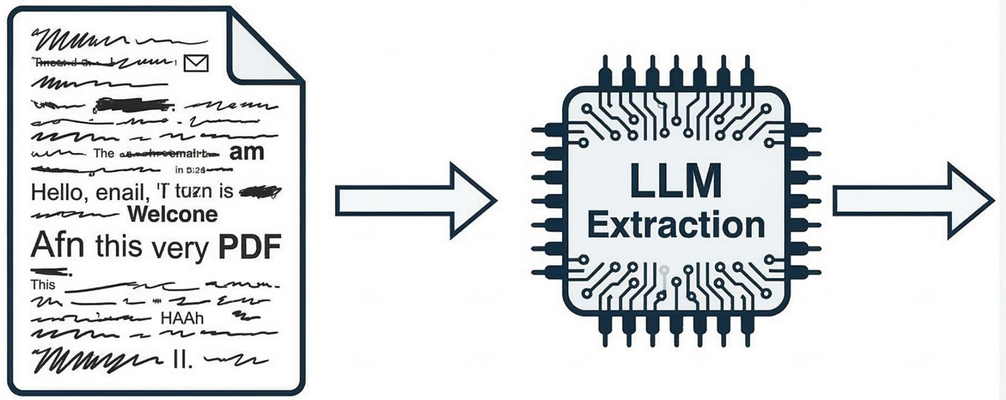
Core Application 1: Advanced Text Summarization

- **The Evolution of Summarization:**
  - *Pre-LLM Era:* Extractive summarization (copying sentences) via smaller encoder models (like BERT).
  - *LLM Era:* Highly abstractive summarization, capable of synthesizing meaning.
- **Key Application Patterns:**
  - **Meeting Transcripts:** Converting raw speech-to-text logs into structured action items.
  - **Document Routing:** Summarizing customer support tickets to categorize and route to the correct human agent.
- **Evaluation Connection:**
  - How do we know the summary is good? As discussed in Week 6, we use metrics like ROUGE, but increasingly rely on **LLM-as-a-judge** (using strict evaluation prompts) to check for *factual consistency* and prevent hallucination.

Core Application 2: Structured Data Extraction

- **The Problem:** 80% of enterprise data is unstructured (emails, PDFs, reports).
- **The LLM Solution:** Using LLMs for Zero-Shot or Few-Shot Information Extraction (IE).
  - **Named Entity Recognition (NER):** Extracting companies, monetary values, and dates without training custom BERT models.
  - **Relation Extraction:** Identifying the relationship between entities (e.g., *Company A [ACQUIRED] Company B*).
- **Implementation Strategy:**
  - Use structural prompting (e.g., instructing the model to output strict JSON).
  - *Challenge:* Output parsing errors. Models may append conversational text ("Here is your JSON:") which breaks downstream code.

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[
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  "subject": "Meeting Update",
  "date": "2023-10-27",
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    },
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      "date": "Meeting"
    }
  }
}
```

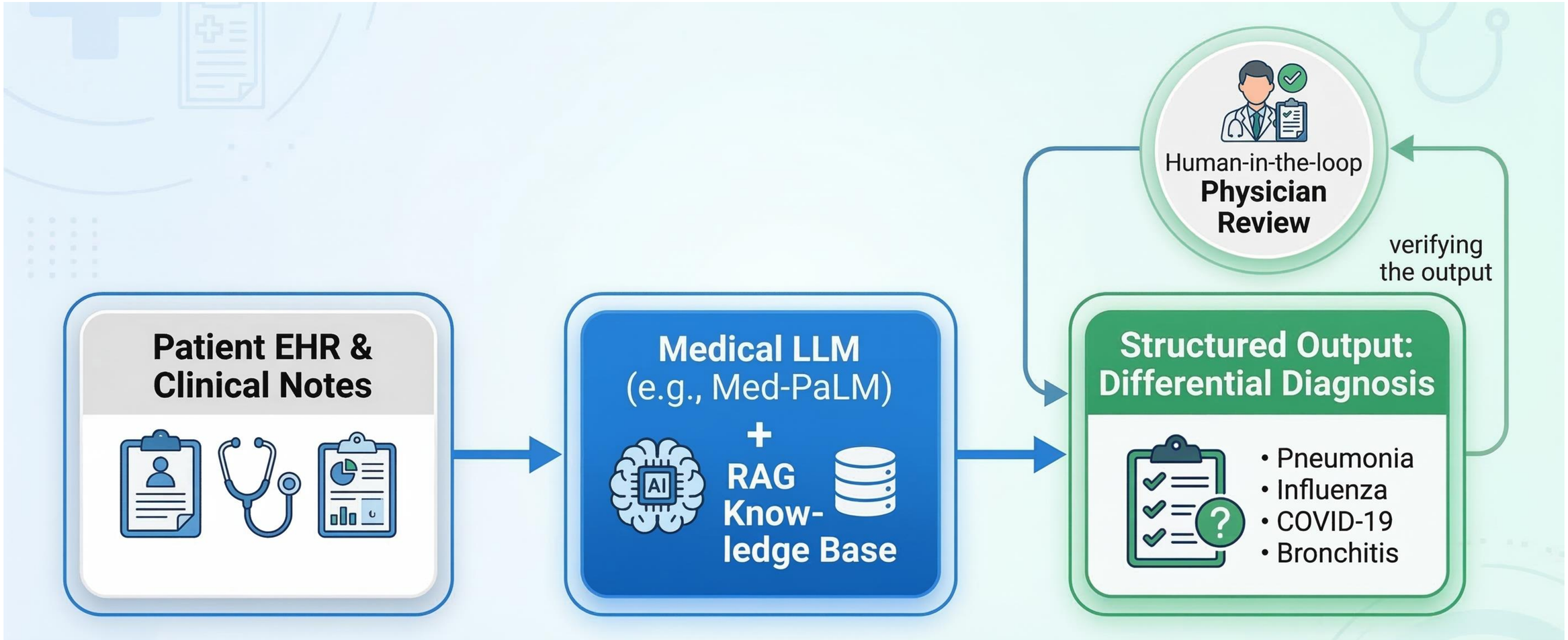
Mini-Quiz 1: Application Architecture

- **Scenario:** You are building an internal company tool to help HR instantly answer employee questions about the 200-page company benefits handbook.
- **Which architecture is the most appropriate and cost-effective?**
  - **A)** Pretrain a new foundational model from scratch on HR documents.
  - **B)** Use a general Instruction-Tuned LLM and Retrieval-Augmented Generation (RAG) with the handbook.
  - **C)** Use a standard Zero-Shot Prompt asking the model to guess the company policies.
  - **D)** Supervised Fine-Tune (SFT) an LLM solely on the HR handbook.

Domain-Specific Application: Software Engineering

- **Code Generation and Copilots:**
  - LLMs trained heavily on GitHub/StackOverflow (e.g., GitHub Copilot, CodeLlama).
  - *Tasks:* Autocompletion, writing unit tests, translating code between languages (e.g., Python to Rust).
- **Code Review and Bug Fixing:**
  - Using LLMs to detect security vulnerabilities or suggest performance optimizations.
- **Why LLMs excel at Code:**
  - Code has strict syntax and logic, reducing the ambiguity present in natural language.
  - *Evaluation:* Extremely objective. We don't need human graders; we evaluate generated code by simply running it through compiler/unit test suites (Pass@k metric).

Domain-Specific Application: Healthcare & Biomedicine



Class Discussion (5 Minutes)

- **Topic: The Generalist vs. Specialist Debate**
- Currently, state-of-the-art general models (like GPT-4) often outperform smaller, domain-specific fine-tuned models (like a 7B parameter legal-specific model) even on domain-specific tasks.
- **Discussion Question:** * Will the future of LLM applications be dominated by a few massive "God-models" accessed via API, or by thousands of smaller, locally hosted, highly specialized models?
  - *Hint: Consider data privacy, inference costs, and latency.*

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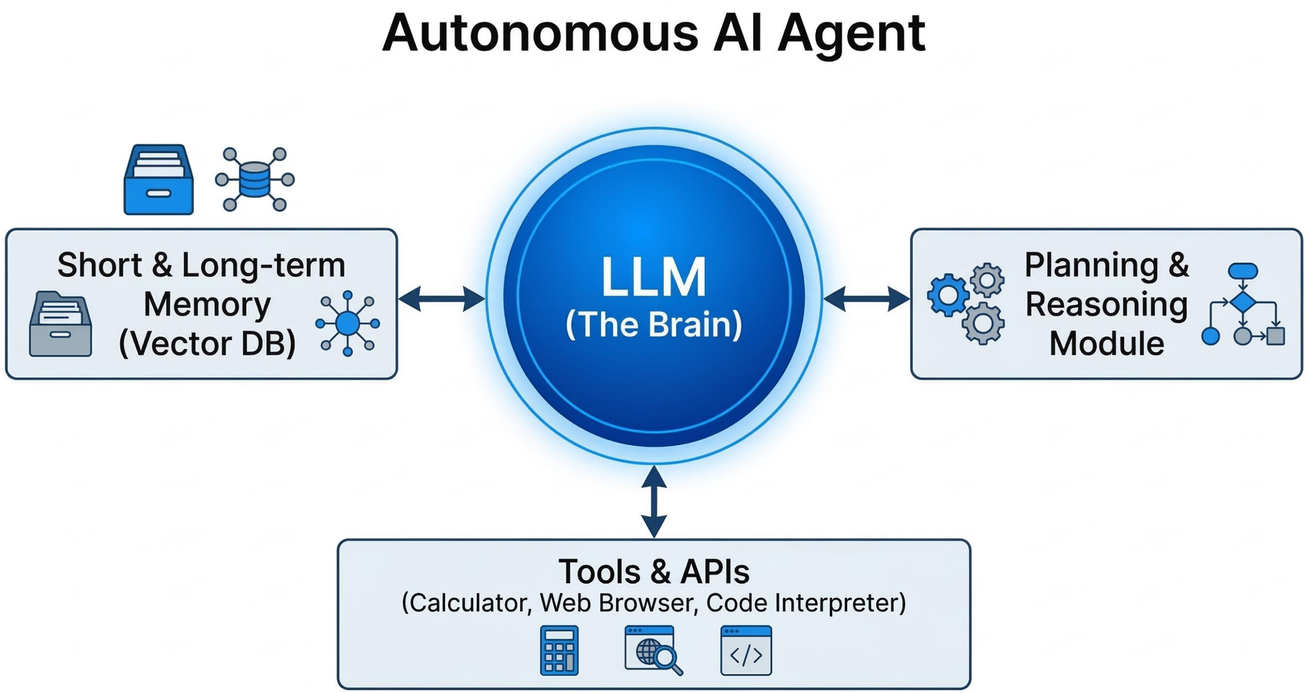
Domain-Specific Application: Law and Finance

- **Legal Sector:**
  - *E-Discovery:* Scanning millions of documents for relevance to a subpoena.
  - *Contract Analysis:* Identifying risky clauses, non-standard terms, or missing indemnifications.
  - *Risk:* The "hallucinated legal cases" problem (models inventing fake precedent).
- **Finance Sector:**
  - *Algorithmic Trading:* Real-time sentiment analysis of financial news and SEC filings.
  - *Automated Reporting:* Synthesizing quarterly earnings calls into executive summaries.
  - *Adaptation technique:* Often heavily relies on RAG to ensure real-time data access (stock prices change by the second; parametric memory is useless here).

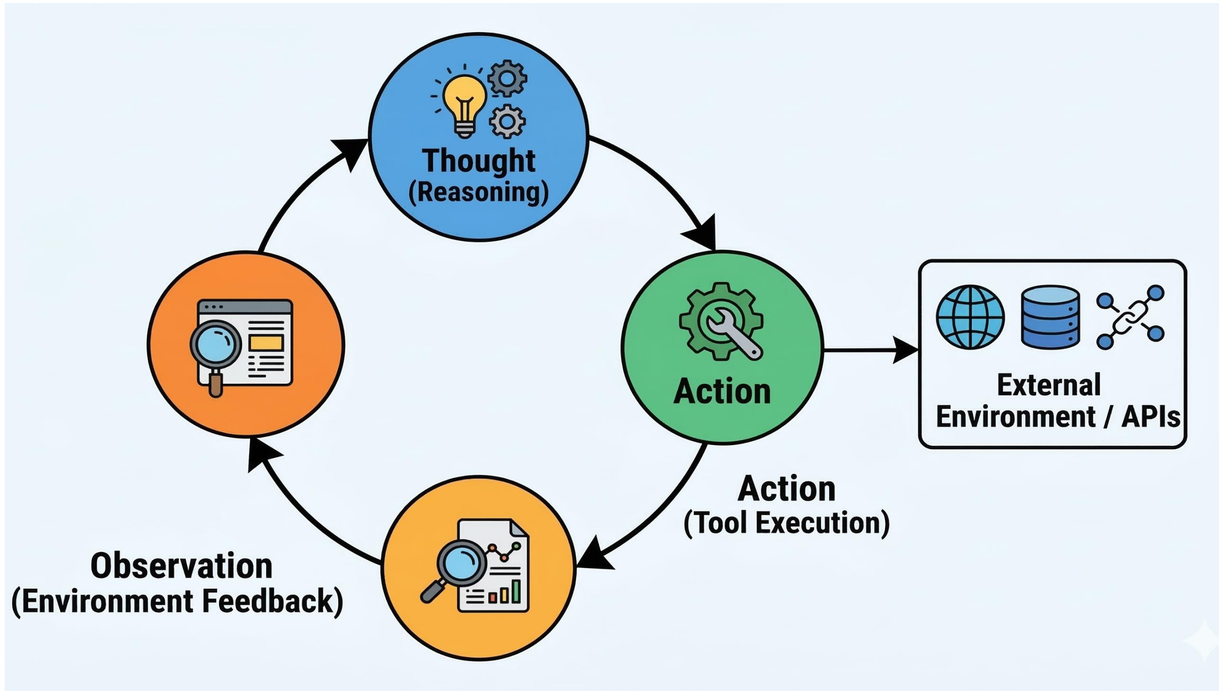
Application Paradigm Shift: LLMs as Agents

- **From Passive Responders to Active Agents:**
  - Standard LLM: Takes text, outputs text.
  - Agentic LLM: Takes a high-level goal, reasons about steps, interacts with external environments, and executes actions.
- **The Core Components of an LLM Agent:**
  - **1. Brain:** The LLM itself (handling reasoning and planning).
  - **2. Memory:** Short-term (context window) an Long-term (vector databases).
  - **3. Tools:** Functions the LLM can call (APIs, calculators, web browsers).



How Agents Work: ReAct Prompting

- **ReAct (Reasoning and Acting):**
  - A specific prompting paradigm that intertwines Chain-of-Thought reasoning with environmental actions.
- **The Loop:**
  - **Thought:** Model reasons about what to do next based on the user prompt.
  - **Action:** Model decides to call an external tool (e.g., SearchWikipedia("Eiffel Tower")).
  - **Observation:** The tool returns data to the model.
  - *(Loop repeats until the goal is achieved).*



- **Why this is powerful:** It overcomes the LLM's inability to do math or know current events by allowing it to use a calculator and a search engine.

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Multi-Agent Systems

- **The Concept:**
  - Instead of one massive prompt trying to do everything, instantiate multiple LLM agents with distinct personas and tools, and let them communicate.
- **Example System (Software Development):**
  - *Agent 1 (Product Manager):* Writes the system requirements.
  - *Agent 2 (Software Engineer):* Writes the Python code based on Agent 1's requirements.
  - *Agent 3 (QA Tester):* Reviews Agent 2's code, finds bugs, and sends it back for revision.
- **Benefits:**
  - Separation of concerns.
  - Self-reflection and self-correction through simulated debate.

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Mini-Quiz 2: Agents and Hallucinations

- **Question:** You ask an LLM Agent to calculate $345,678 \times 987,654$. Which of the following describes the safest "Agentic" approach?
- **A)** Prompt the LLM with "Let's think step-by-step" to calculate the math token-by-token.
- **B)** Provide the LLM with a Python interpreter tool, instruct it to write a Python script to multiply the numbers, execute it, and return the printed observation.
- **C)** Fine-tune the LLM on thousands of multiplication tables.

Deployment Bottlenecks: Why isn't LLM everywhere yet?

- Despite the amazing capabilities, real-world deployment faces severe engineering hurdles:
- **1. Inference Cost:**
  - Running a 70B parameter model costs significant GPU compute per token. Generative AI is structurally more expensive than traditional software.
- **2. Latency:**
  - Autoregressive generation (token-by-token) is slow. High latency degrades user experience in real-time applications (like voice assistants).
- **3. Reliability & Evaluation:**
  - Traditional software is deterministic (Input A always yields Output B).
  - LLMs are probabilistic. Continuous monitoring and robust evaluation pipelines (like the Inversion Prompting techniques covered in Week 6) are strictly required.

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Ethical and Safety Considerations in Applications

- **Bias and Fairness:**
  - Models amplify biases present in their training data (e.g., screening resumes and unfairly penalizing certain demographics).
- **Security (Prompt Injection):**
  - User-facing applications (like customer service bots) are highly vulnerable to malicious users altering the system prompt to leak data or output profanity.
- **Copyright and IP:**
  - Generative applications risk reproducing copyrighted code or text from their pretraining datasets.

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Discussion: The Future of Applications

- **Topic: Human-in-the-loop vs. Full Autonomy**
- Currently, most successful LLM applications act as "Copilots" (augmenting a human worker who reviews the final output).
- However, the industry is pushing toward "Agents" (full autonomy, executing workflows while humans sleep).
- **Discussion Prompt:**
  - In which industries is full LLM autonomy acceptable today?
  - Where must we permanently maintain a "human-in-the-loop" constraint?

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Summary & Q&A

- **Summary:**
  - LLM Applications extend base models using Prompting, RAG, and Tool-use to solve real business problems.
  - Core NLP tasks (Summarization, Extraction) are now largely solved by prompting.
  - Domain applications (Code, Medical, Legal) require strict adherence to facts, often leveraging RAG.
  - **Agents** represent the frontier: models that use tools, reason via ReAct, and interact with the environment.
  - Deployment is currently constrained by compute costs, latency, and safety evaluation.

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Multimodal LLMs

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Agenda & Learning Outcomes

- **Agenda:**
  - The Motivation for Multimodality
  - Core MLLM Architecture & The "Modality Gap"
  - Landmark Models (CLIP, Flamingo, BLIP-2, LLaVA)
  - Training Pipelines: Visual Instruction Tuning
  - Multimodal Prompting & Evaluation
- **Learning Outcomes:**
  - **Understand** the structural components of an MLLM (Encoder, Projector, LLM Backbone).
  - **Analyze** the two-stage training pipeline for modern vision-language models.
  - **Apply** multimodal prompting techniques (interleaved text/image).
  - **Evaluate** MLLMs using modern benchmarks and identify visual hallucination risks.

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What is a Multimodal LLM (MLLM)?

- **Definition:** * An MLLM is a language model capable of receiving, reasoning over, and (sometimes) generating multiple data modalities (text, images, audio, video).
- **The Goal:**
  - Retain the advanced reasoning and instruction-following capabilities of the LLM.
  - Grant the LLM "eyes" and "ears" to perceive the real world.
- **Focus of this Lecture:**
  - Vision-Language Models (VLMs), as they are currently the most researched and deployed subclass of MLLMs.

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Core Architecture of an MLLM

- Virtually all modern MLLMs share a tripartite (three-part) architecture:
- **1. The Vision Encoder:**
  - Extracts high-dimensional feature representations from raw image pixels.
  - *Example:* Vision Transformer (ViT).
- **2. The Connector (Projection Layer):**
  - Translates the visual features into the "language" (embedding space) of the LLM.
- **3. The LLM Backbone:**
  - The brain. Processes the translated visual tokens alongside standard text tokens to generate a text response.
  - *Example:* LLaMA, Vicuna.

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Component 1: The Vision Encoder (CLIP)

- **Contrastive Language-Image Pretraining (CLIP):**
  - A foundational model by OpenAI that maps images and text into the *same* vector space.
- **How it works:**
  - Trained on 400 million (Image, Caption) pairs.
  - Uses a Contrastive Loss function: maximizes the cosine similarity between the embedding of an image (e.g., a dog) and its correct text description ("a picture of a dog").
- **Why it matters for MLLMs:**
  - We freeze the CLIP Vision Encoder and use it as the "eyes" for our LLMs, because its visual features are already semantically aligned with human language.

Reference: Radford et al., "Learning Transferable Visual Models From Natural Language Supervision" (ICML 2021)

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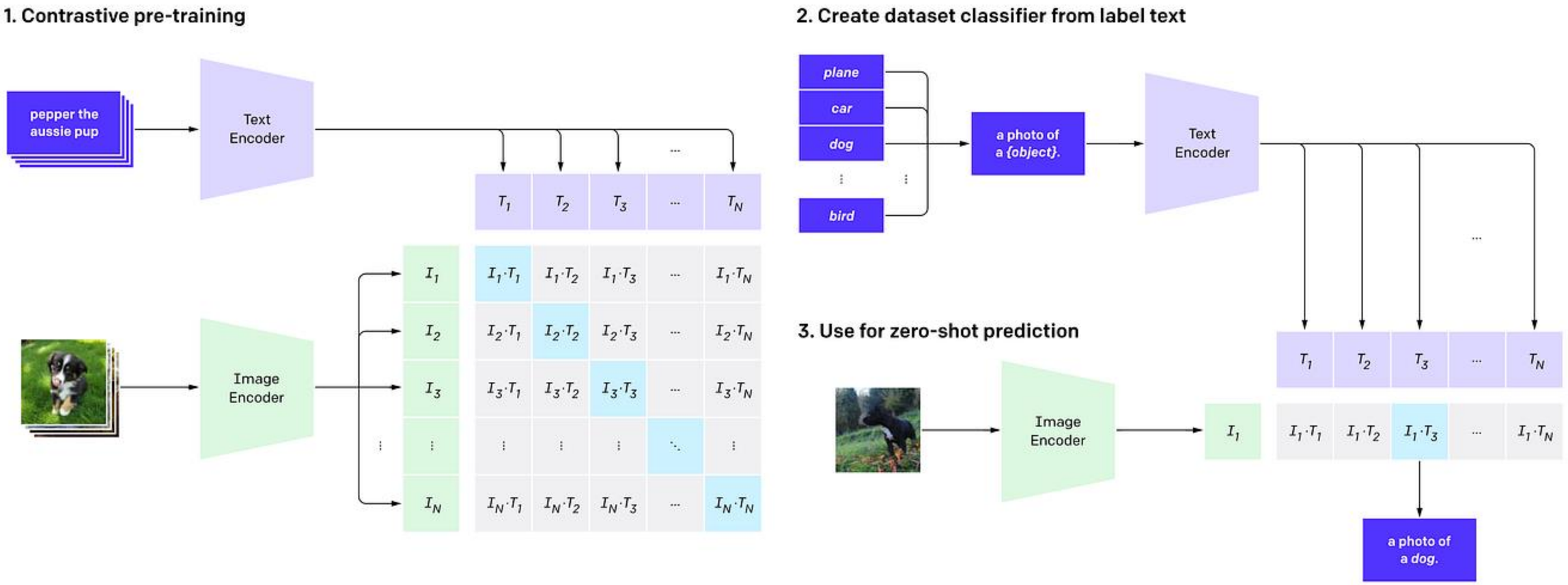
CLIP Architecture Overview



CLIP pre-trains an image encoder and a text encoder to predict which images were paired with which texts in our dataset. We then use this behavior to turn CLIP into a zero-shot classifier. We convert all of a dataset's classes into captions such as "a photo of a dog" and predict the class of the caption CLIP estimates best pairs with a given image.

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Component 2: The Connector / Modality Gap

- **The Problem:**
  - The CLIP Vision Encoder outputs vectors of size D_{vision}.
  - The LLM expects word embeddings of size D_{text}.
  - These two spaces do not naturally communicate (The "Modality Gap").
- **The Solution:**
  - We introduce a lightweight, trainable neural network layer between them.
- **Connector Types:**
  - *Linear Projection:* A simple weight matrix (used in early LLaVA).
  - *MLP (Multi-Layer Perceptron):* A 2-layer network for better translation.
  - *Q-Former / Perceiver Resampler:* Compresses visual tokens to save context window space (used in BLIP-2, Flamingo).

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Mini-Quiz 1: Architecture Check

- **Question:** In a standard vision-language MLLM like LLaVA, which component is primarily responsible for performing complex logic, math, and generating the final conversational response?
- **A)** The Vision Transformer (ViT)
- **B)** The Contrastive Loss Function
- **C)** The LLM Backbone (e.g., Vicuna/LLaMA)
- **D)** The Projection Layer

Landmark Model 1: Flamingo

- **The Breakthrough:** DeepMind's Flamingo was the first model to demonstrate highly effective interleaved image-text prompting.
- **Architecture Innovation:**
  - *Perceiver Resampler:* Compresses thousands of image tokens down to just 64 tokens, saving massive computational cost.
  - *Cross-Attention Layers:* Instead of feeding the image at the very beginning of the prompt, Flamingo injects visual information directly into the deep layers of the LLM.
- **Impact:** Proved that MLLMs could perform Few-Shot In-Context Learning (just like GPT-3), but with images.

Reference: Alayrac et al., "Flamingo: a Visual Language Model for Few-Shot Learning" (NeurIPS 2022)

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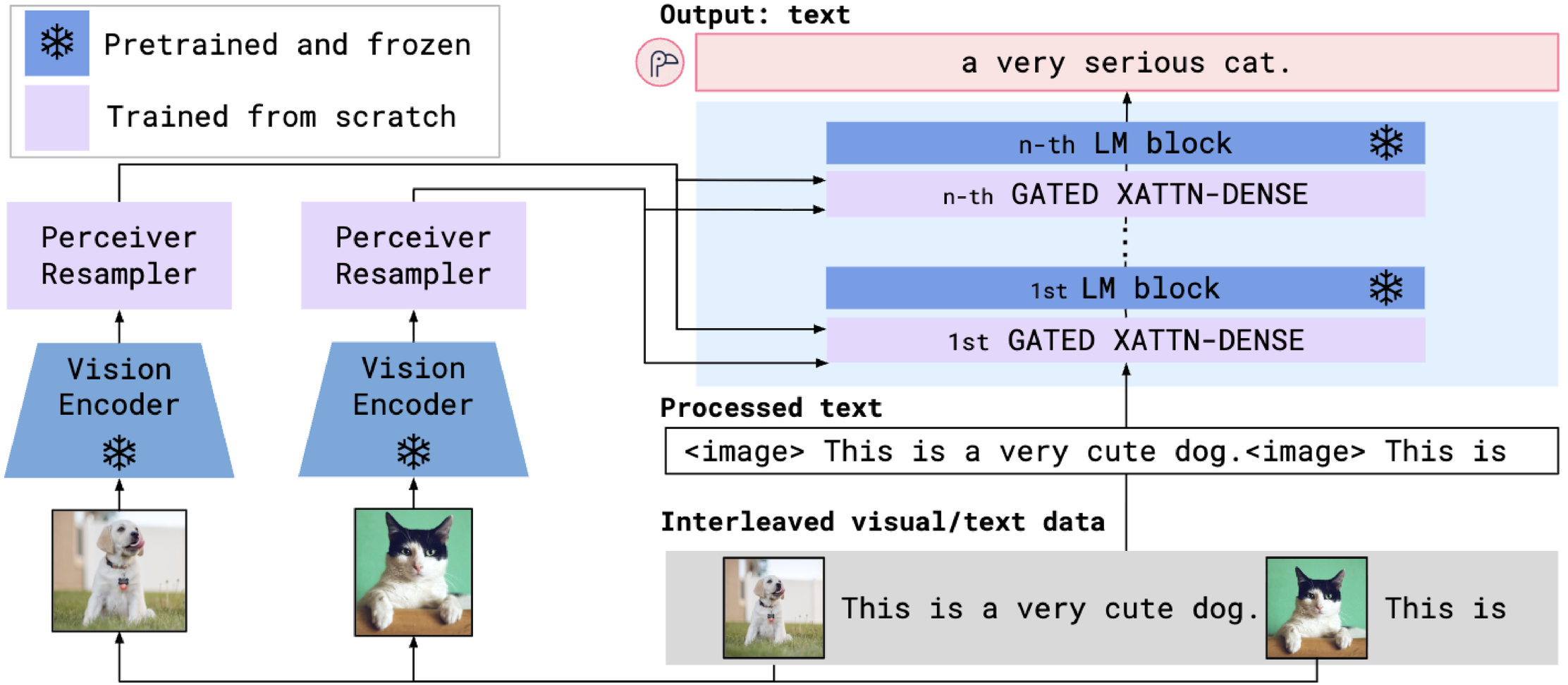
Flamingo Architecture Overview



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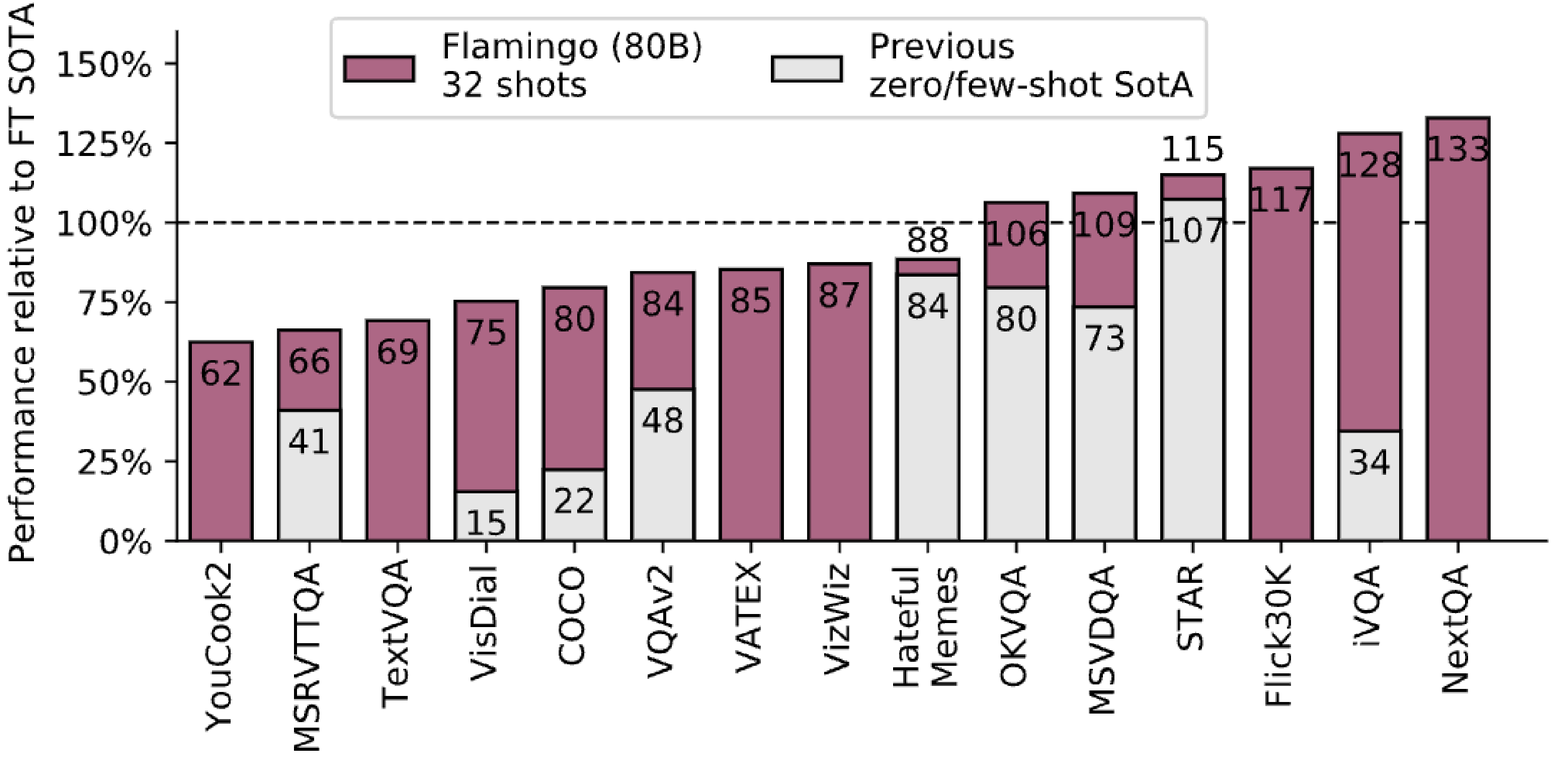
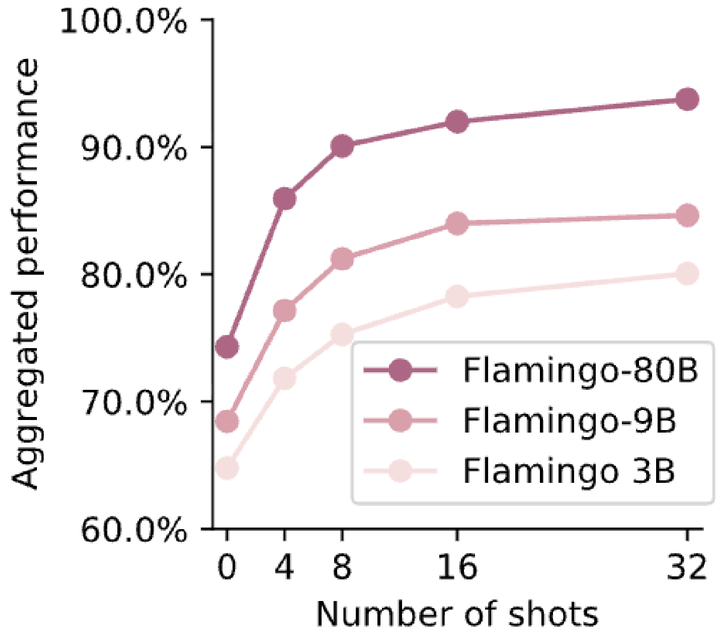
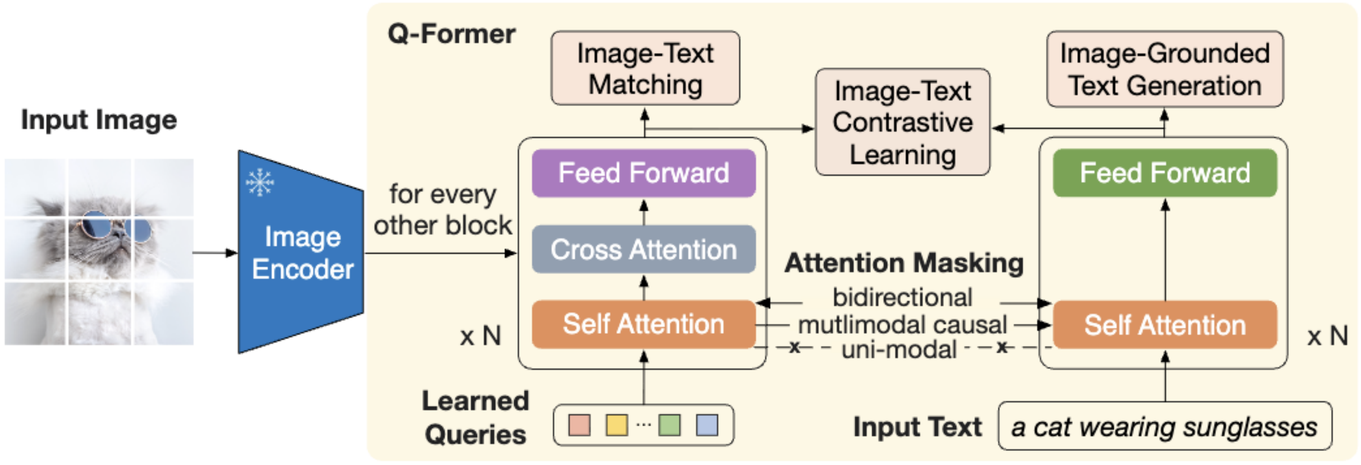
Flamingo Results Overview

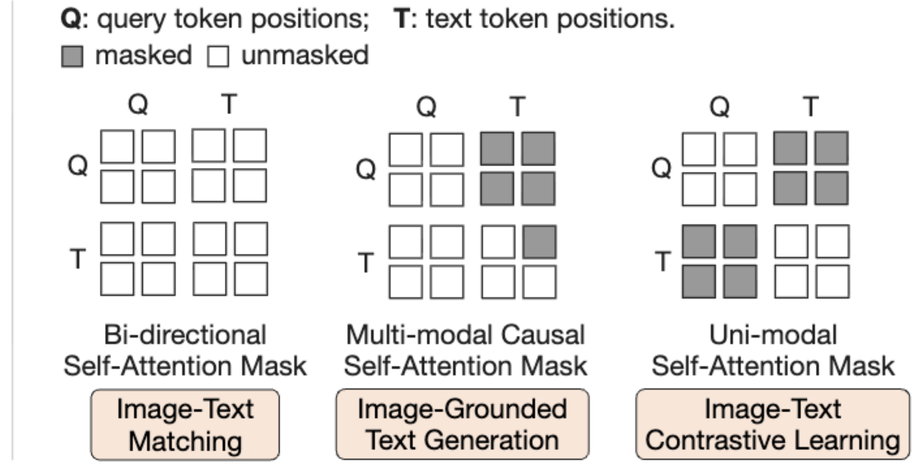


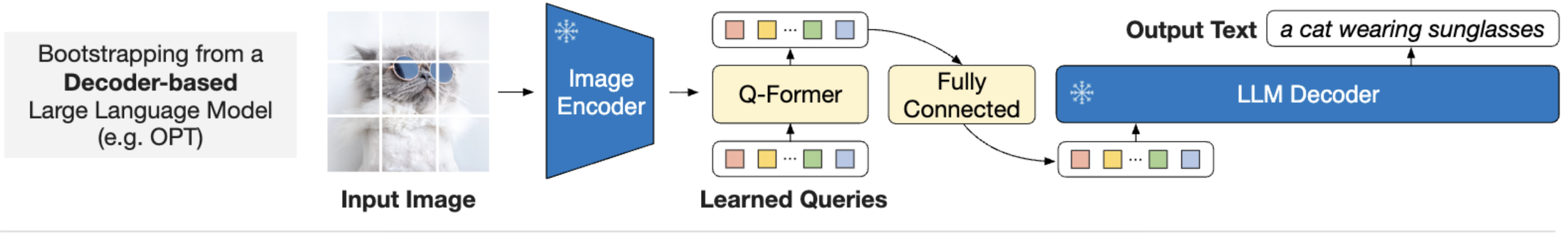


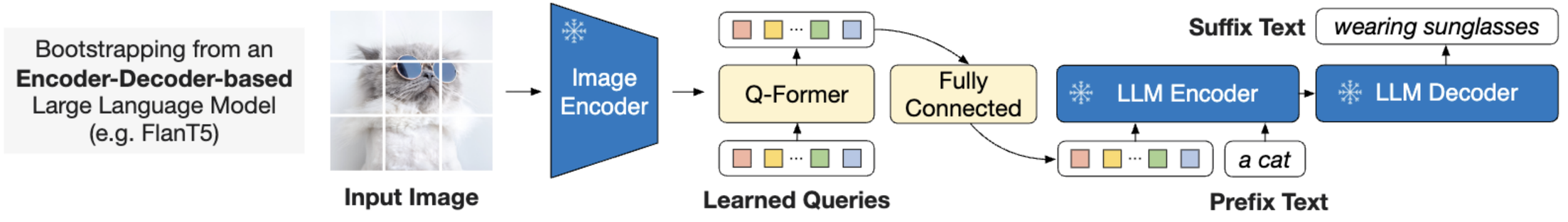
Landmark Model 2: BLIP-2

- *Reference: Li et al., "BLIP-2: Bootstrapping Language-Image Pre-training..." (ICML 2023)*
- **The Efficiency Breakthrough:**
  - Training massive LLMs from scratch is expensive.
  - BLIP-2 completely *freezes* both the Vision Encoder and the LLM Backbone.
- **The Q-Former (Querying Transformer):**
  - A lightweight, trainable module that learns to extract only the most relevant visual features required by the LLM.
- **Result:** Achieved state-of-the-art performance using significantly fewer trainable parameters, democratizing MLLM research for university labs.


Input Image
Image Encoder
Q-Former
Image-Text Matching
Image-Text Contrastive Learning
Image-Grounded Text Generation
for every other block
Feed Forward
Cross Attention
Self Attention
x N
Attention Masking
bidirectional
mutlimodal causal
uni-modal
Feed Forward
Self Attention
x N
Learned Queries
Input Text
a cat wearing sunglasses


Q: query token positions; T: text token positions.
masked unmasked
Q T
Q
T
Bi-directional Self-Attention Mask
Image-Text Matching
Multi-modal Causal Self-Attention Mask
Image-Grounded Text Generation
Uni-modal Self-Attention Mask
Image-Text Contrastive Learning


Bootstrapping from a **Decoder-based** Large Language Model (e.g. OPT)
Input Image
Image Encoder
Learned Queries
Q-Former
Fully Connected
LLM Decoder
Output Text
a cat wearing sunglasses


Bootstrapping from an **Encoder-Decoder-based** Large Language Model (e.g. FlanT5)
Input Image
Image Encoder
Learned Queries
Q-Former
Fully Connected
LLM Encoder
LLM Decoder
Suffix Text
wearing sunglasses
a cat
Prefix Text

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Landmark Model 3: LLaVA

- **Large Language-and-Vision Assistant (LLaVA):**
  - The open-source pioneer of Visual Instruction Tuning.
- **Simplicity is Key:**
  - Uses a frozen CLIP encoder.
  - Uses a simple Linear/MLP projection layer.
  - Uses a frozen Vicuna (LLaMA-based) LLM.
- **Why did it win?** The magic wasn't in the architecture; it was in the *data generation pipeline* (using GPT-4 to generate multimodal training data).

Reference: Liu et al., "Visual Instruction Tuning" (NeurIPS 2023)

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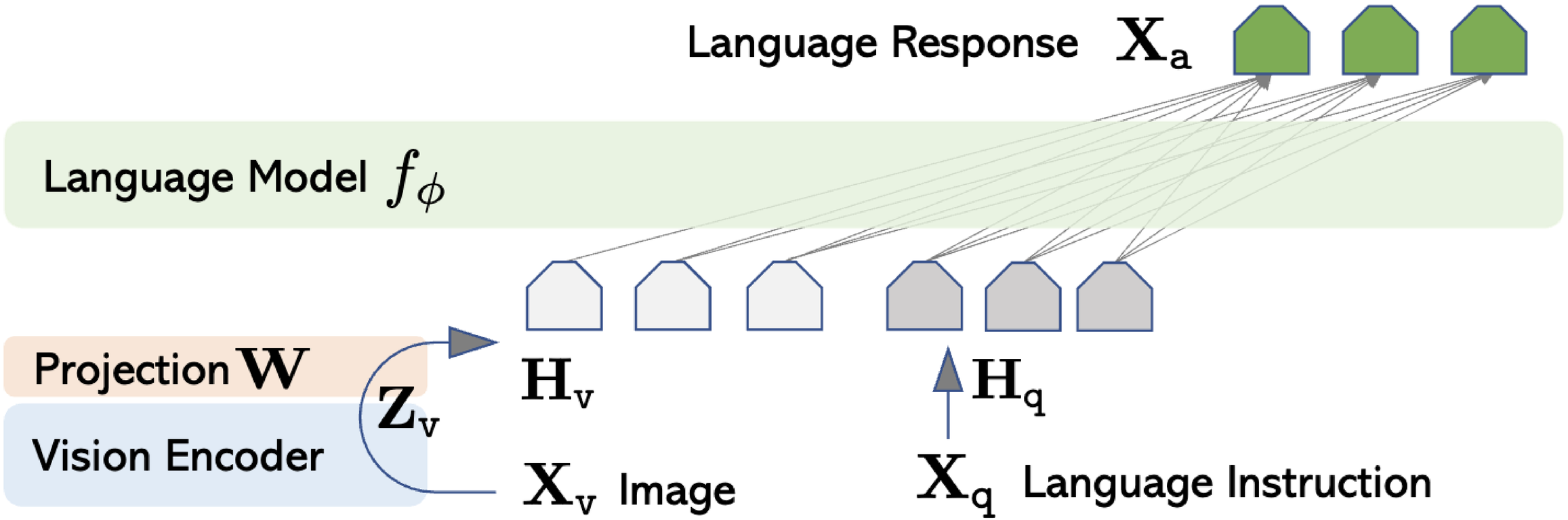
LLaVA Architecture Overview



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The MLLM Training Pipeline (Stage 1)

- Training an MLLM like LLaVA happens in two distinct stages.
- **Stage 1: Feature Alignment Pretraining**
  - **Goal:** Teach the LLM to understand the Vision Encoder's "language".
  - **Data:** Millions of simple (Image, Caption) pairs (e.g., CC3M dataset).
  - **Frozen:** Vision Encoder and LLM Backbone.
  - **Trainable:** *Only* the Projection Layer.
  - **Analogy:** Teaching an English speaker (LLM) to read a French dictionary (Vision features) by mapping words one-to-one.

The MLLM Training Pipeline (Stage 2)

- **Stage 2: Visual Instruction Tuning (Multimodal SFT)**
  - **Goal:** Teach the model to follow complex human instructions about images (e.g., "Why is this meme funny?").
  - **Data:** ~150k highly detailed, instruction-following multimodal conversations.
  - **Frozen:** Vision Encoder.
  - **Trainable:** Projection Layer *and* LLM Backbone.
- **Connecting to Week 6:** Just like standard Supervised Fine-Tuning (SFT) taught text LLMs to be assistants, Visual Instruction Tuning teaches MLLMs to be *visual* assistants.

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Class Discussion: Data Generation (5 mins)

- **Topic: How do we get Visual Instruction Data?**
- In 2023, the LLaVA authors needed 150,000 conversational Q&A pairs about images, but human labeling was too expensive.
- They used a text-only model (GPT-4) to generate data for their multimodal model.
- **Discussion Question:**
  - If GPT-4 (at the time) was text-only and couldn't see images, how did the authors use it to generate highly accurate, detailed visual Q&A data to train LLaVA?
  - *(Hint: Think about bounding boxes and captions).*

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Beyond Images: Audio and Video

- **Video as a Sequence of Images:**
  - Video MLLMs (like Video-LLaVA) treat video as a sequence of frames.
  - *Challenge:* Context window exhaustion. 30 frames of video = 30x the visual tokens.
- **Audio Encoders:**
  - Architectures like Whisper (Audio Transformer) can be projected into the LLM just like CLIP.
- **Any-to-Any Models:**
  - Emerging models (like GPT-4o or Gemini 1.5 Pro) are trained natively on multiple modalities from scratch, bypassing the strict "Encoder -> Projector" pipeline for more organic cross-modal understanding.

Evaluating MLLMs: Traditional Benchmarks

- How do we know if an MLLM is good?
- **1. VQAv2 (Visual Question Answering):**
  - Short, factual questions (e.g., "What color is the car?").
  - *Drawback:* Too simple for modern MLLMs.
- **2. OK-VQA (Outside Knowledge VQA):**
  - Requires world knowledge not present in the image.
  - *Example:* `` + "Which political party does this person belong to?"
- **3. Image Captioning (COCO):**
  - Generating descriptive text. Evaluated via n-gram overlap metrics like CIDEr or BLEU.

Evaluating MLLMs: Comprehensive Benchmarks

- Because modern MLLMs are conversational, old benchmarks fail to capture their abilities.
- **1. MME (Multimodal Evaluation):**
  - Tests perception (color, position, count) and cognition (commonsense, math, code).
  - Evaluates robustness using Yes/No prompts.
- **2. MM-Vet:**
  - Evaluates complex, multi-turn multimodal dialogue.
  - Uses **LLM-as-a-Judge** (usually GPT-4V) to grade the MLLM's responses based on accuracy, helpfulness, and reasoning.

Mini-Quiz 2: Evaluation Constraints

- **Scenario:** You evaluate your new MLLM on VQAv2.
- Ground Truth Answer: "Red"
- Your Model's Answer: "The car shown in the image is red."
- **Question:** What happens to your evaluation score under strict traditional string-matching metrics (like Exact Match)?
  - **A)** Score is 100% (Correct).
  - **B)** Score is 0% (Incorrect).
- **Answer: B.** * *Insight:* Instruction-tuned MLLMs are "chatty". Traditional benchmarks penalize them for full sentences. This is why LLM-as-a-judge is now the standard for MLLM evaluation.

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The Hallucination Problem in MLLMs

- **Visual Hallucination:** * When an MLLM confidently describes objects, attributes, or relationships that do *not* exist in the provided image.
- **Why does it happen?**
  - **Language Prior:** The LLM backbone is so powerful that it overrides the visual encoder. (e.g., It sees a dining table and hallucinates "chairs" because textually, chairs usually accompany tables).
  - **Connector Compression:** Small details get lost when reducing megapixel images to a few hundred tokens.

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Evaluating Hallucinations: POPE

- **Polling-based Object Probing Evaluation (POPE):**
  - A benchmark specifically designed to expose visual hallucinations.
- **Methodology:**
  - Frames evaluation as binary Yes/No questions.
  - *Adversarial Probing:* Asks about objects that frequently co-occur but are missing from the specific image (e.g., `` -> "Is there a mouse in the image?").
- **Finding:** Many early MLLMs failed POPE catastrophically, defaulting to "Yes" due to statistical priors.

Reference: Li et al., "Evaluating Object Hallucination in Large Vision-Language Models" (EMNLP 2023)

Class Discussion: The "Clever Hans" Effect

- *Context:* Clever Hans was a horse who appeared to do math, but was actually just reading the subtle body language of his trainer.
- **Discussion Question:**
  - Are MLLMs truly "seeing" the image, or are they experiencing a Clever Hans effect where the user's prompt gives away the answer?
  - *Example:* If I prompt: "What is the man holding in his left hand?", does the model know there is a man, or does it just assume there is based on my question? How can we design prompts to prevent this?